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import java.util.Scanner;

public class ArrayOperations_YourIndexNumber {

    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        // 1. Create an integer array with user input
        System.out.print("Enter the number of elements (n): ");
        int n = input.nextInt();
        int[] arr = new int[n];

        System.out.println("Enter " + n + " integers:");
        for (int i = 0; i < n; i++) {
            arr[i] = input.nextInt();
        }
        printArray("Initial Array", arr, n);

        // 2. Delete an element from a given position
        System.out.print("\nEnter position to delete (0 to " + (n - 1) + "): ");
        int delPos = input.nextInt();

        if (delPos >= 0 && delPos < n) {
            // Shift elements to the left to fill the gap
            for (int i = delPos; i < n - 1; i++) {
                arr[i] = arr[i + 1];
            }
            n--; // Decrease size after deletion
            printArray("After Deletion", arr, n);
        } else {
            System.out.println("Invalid position! No deletion performed.");
        }

        // 3. Update an element at a given position
        System.out.print("\nEnter position to update (0 to " + (n - 1) + "): ");
        int updPos = input.nextInt();

        if (updPos >= 0 && updPos < n) {

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        System.out.print("Enter new value: ");
        int newValue = input.nextInt();
        arr[updPos] = newValue;
        printArray("After Update", arr, n);
    } else {
        System.out.println("Invalid position! No update performed.");
    }

    // 4. Search for a value
    System.out.print("\nEnter value to search for: ");
    int searchVal = input.nextInt();
    int foundIndex = -1;

    for (int i = 0; i < n; i++) {
        if (arr[i] == searchVal) {
            foundIndex = i;
            break; // Stop once found
        }
    }

    if (foundIndex != -1) {
        System.out.println("Element found at index: " + foundIndex);
    } else {
        System.out.println("Element not found");
    }

    input.close();
}

// 5. Helper method to display the array
public static void printArray(String label, int[] array, int size) {
    System.out.print(label + ": ");
    for (int i = 0; i < size; i++) {
        System.out.print(array[i] + " ");
    }
    System.out.println();
}
}

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